What Statistic Predicts MLB Success?



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Agenda



Research Question

Our Data

Difference in Means

Regression Analysis

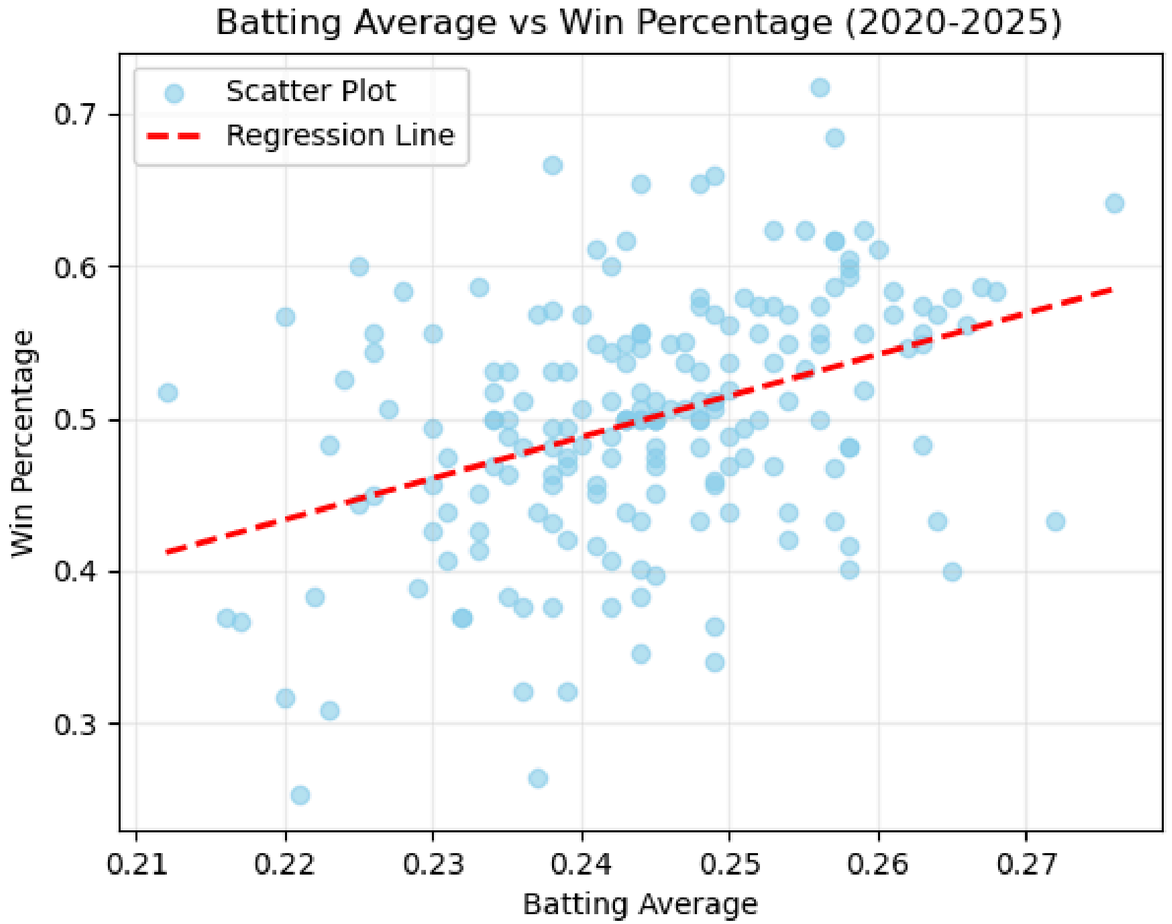
Conclusion

Research Question

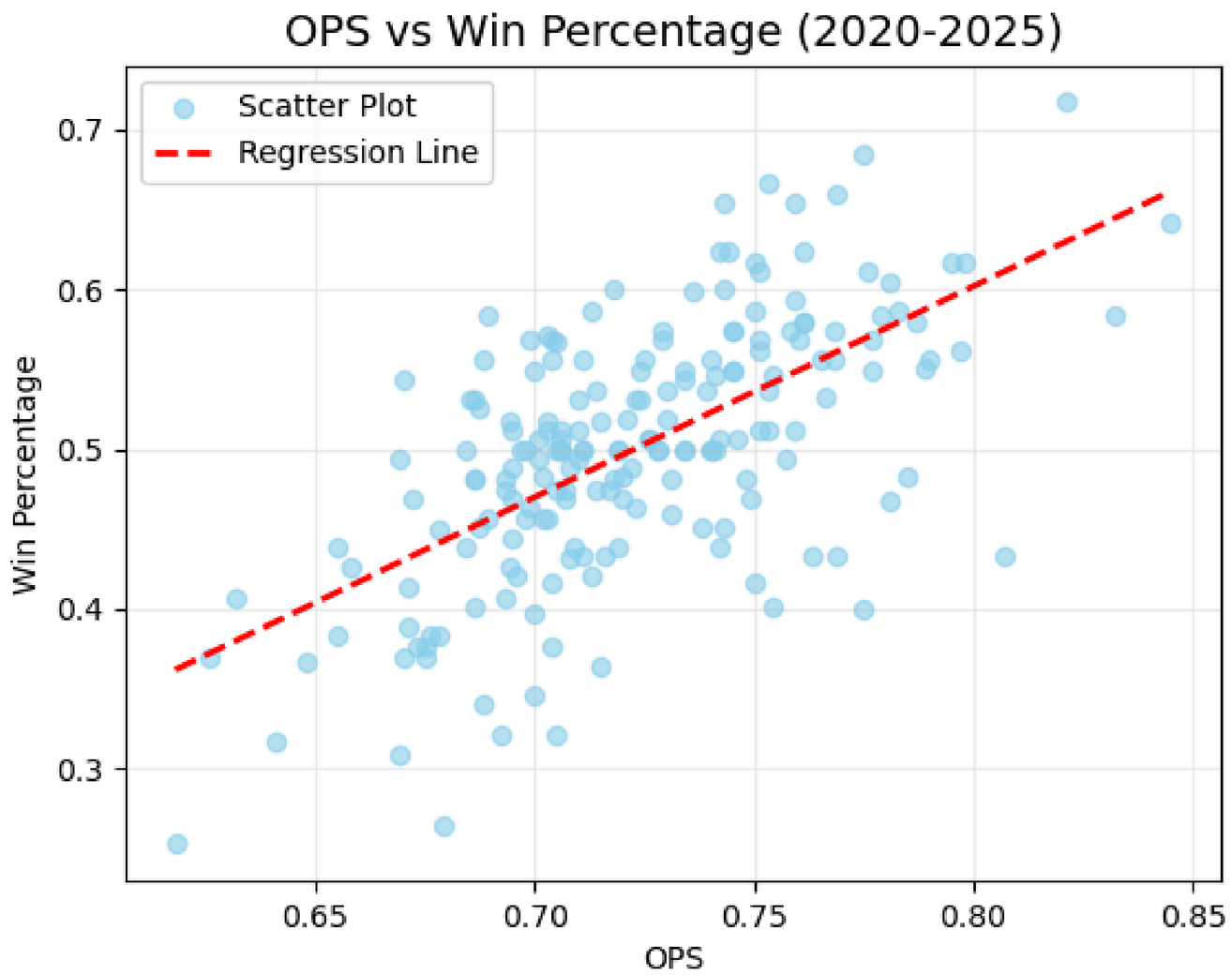
Is BA, OPS, or ERA more strongly associated with team success in the MLB from 2020 to 2025?

Batting Average vs Win Percentage

- Relationship between batting average and winning
- Positive slope suggests higher BA leads to more wins
- Data points are widely spread
- Indicates weak relationship

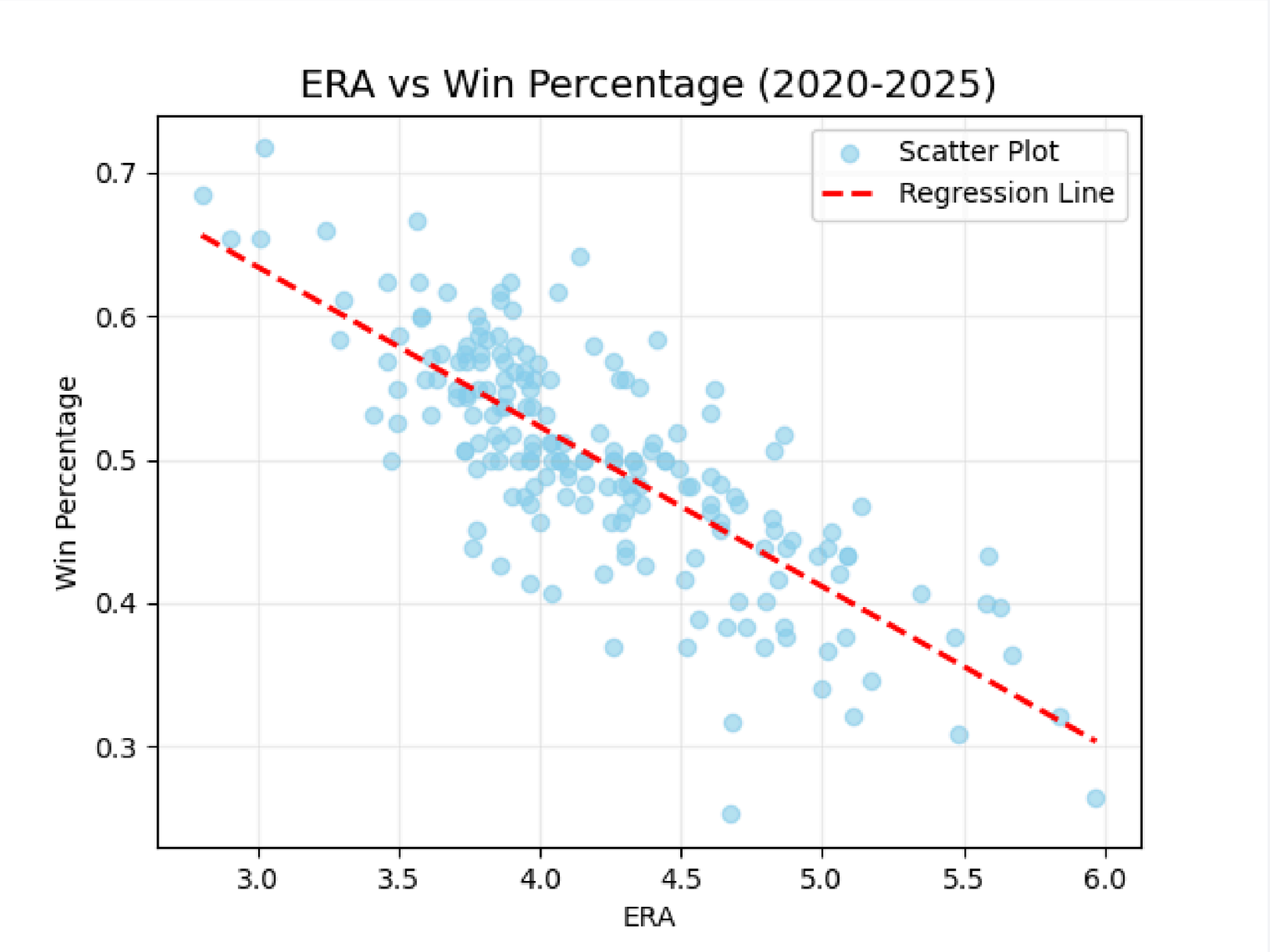


OPS vs Win Percentage



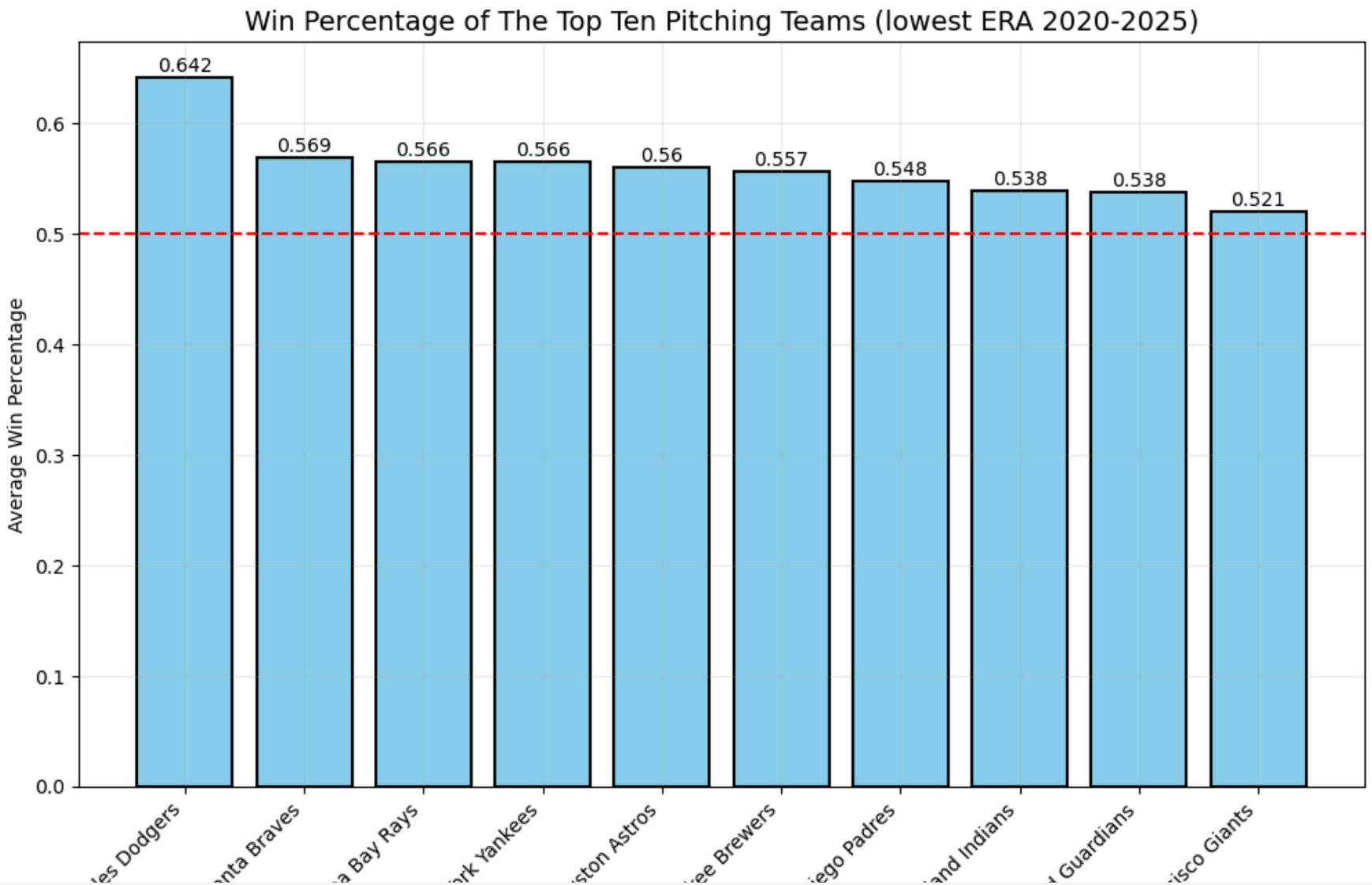
- OPS includes on-base percentage and power
- Shows stronger relationship than batting average
- Points are more tightly clustered
- Suggests better offensive predictor

ERA vs Win Percentage



- ERA measures pitching performance
- Lower ERA = better pitching
- Negative slope is good here
- More wins with lower ERA
- Strongest visual relationship
- Suggests pitching has major impact on team success

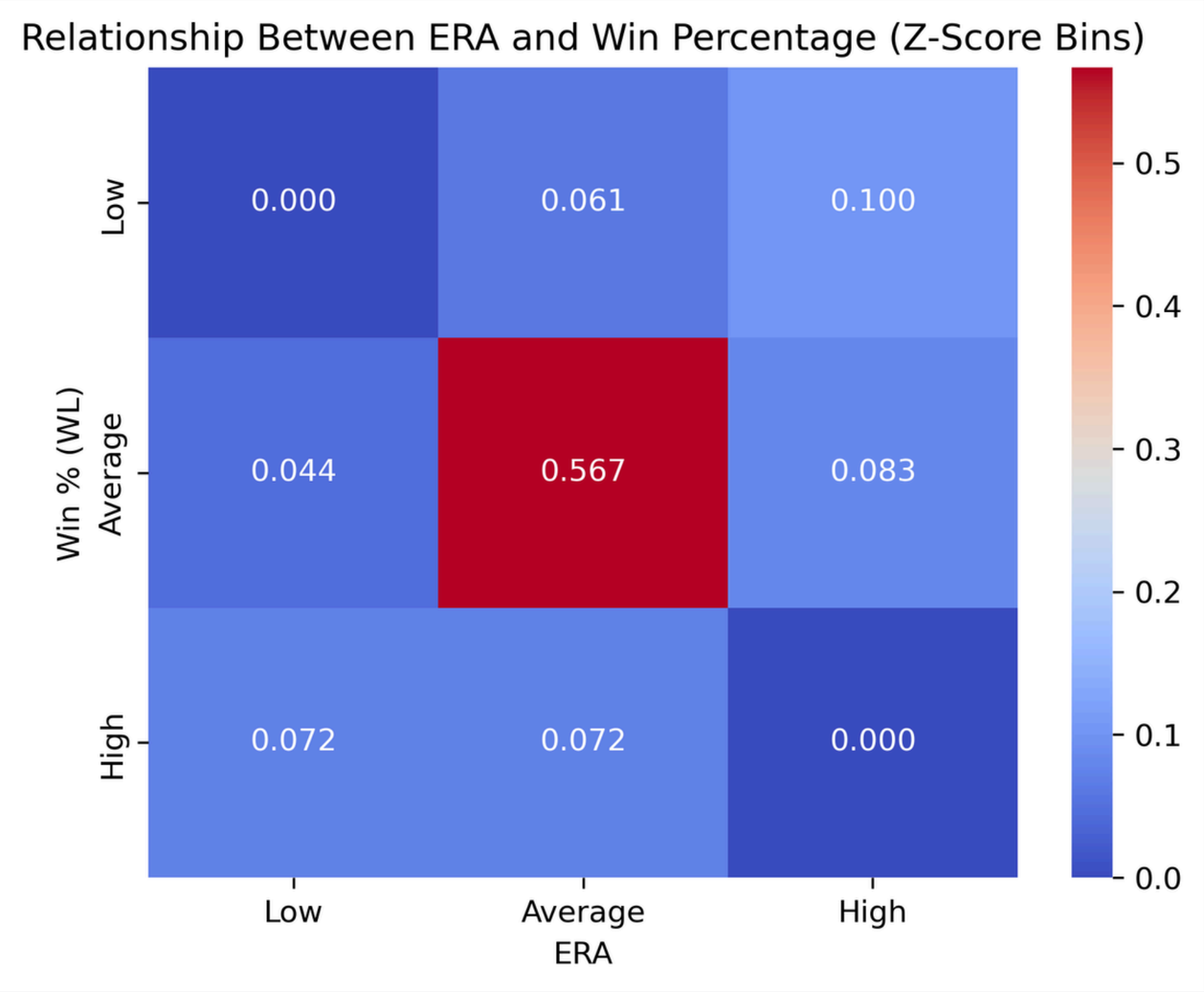
Win Percentage of Top Ten Pitching Teams



- Shows teams with lowest avg ERA from 2020-2025
- Displays their average winning percentage
- All top ten pitching teams are above 50% win rate
- Reinforces importance of strong pitching

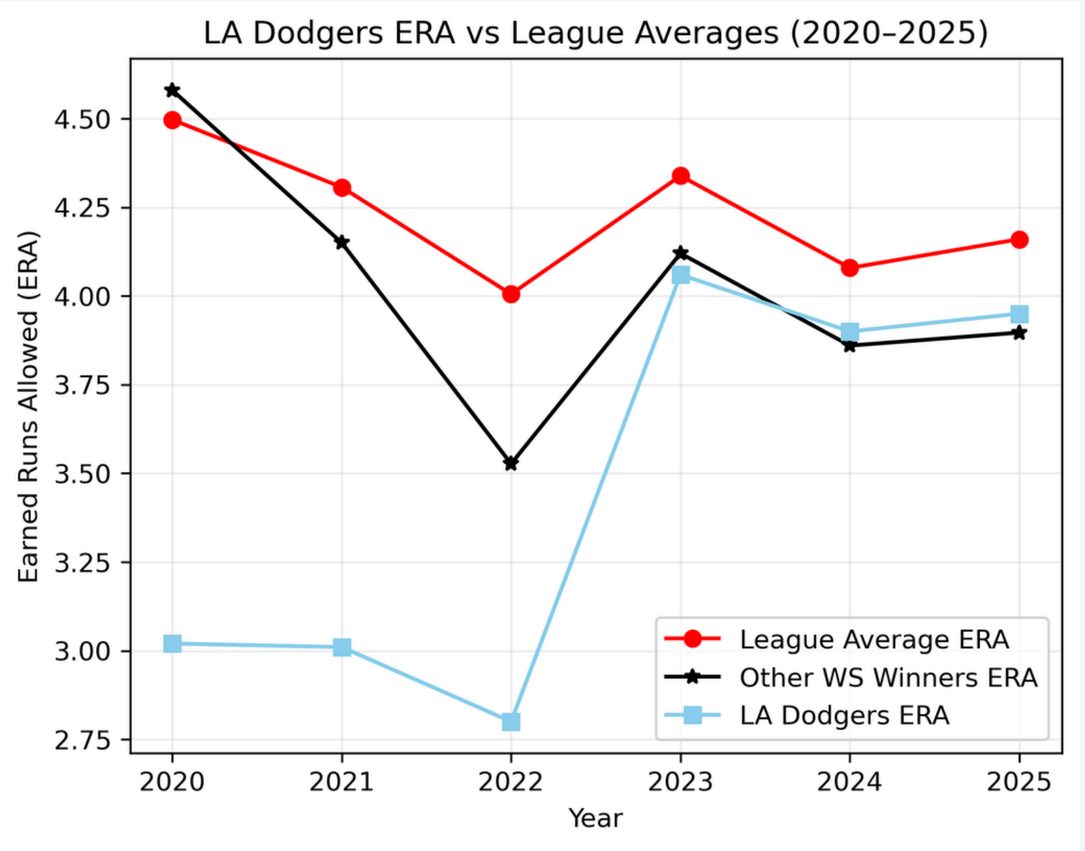
Relationship Between ERA & WL%

- This is an inverse relationship
- High = >+1 STD
- Average = between ±1 STD
- Low = <-1 STD
- Key things:
  - Both average: includes 56.7% of teams
  - Both high/low: teams don't win
  - ~70% of teams fall into the average ERA & WL%



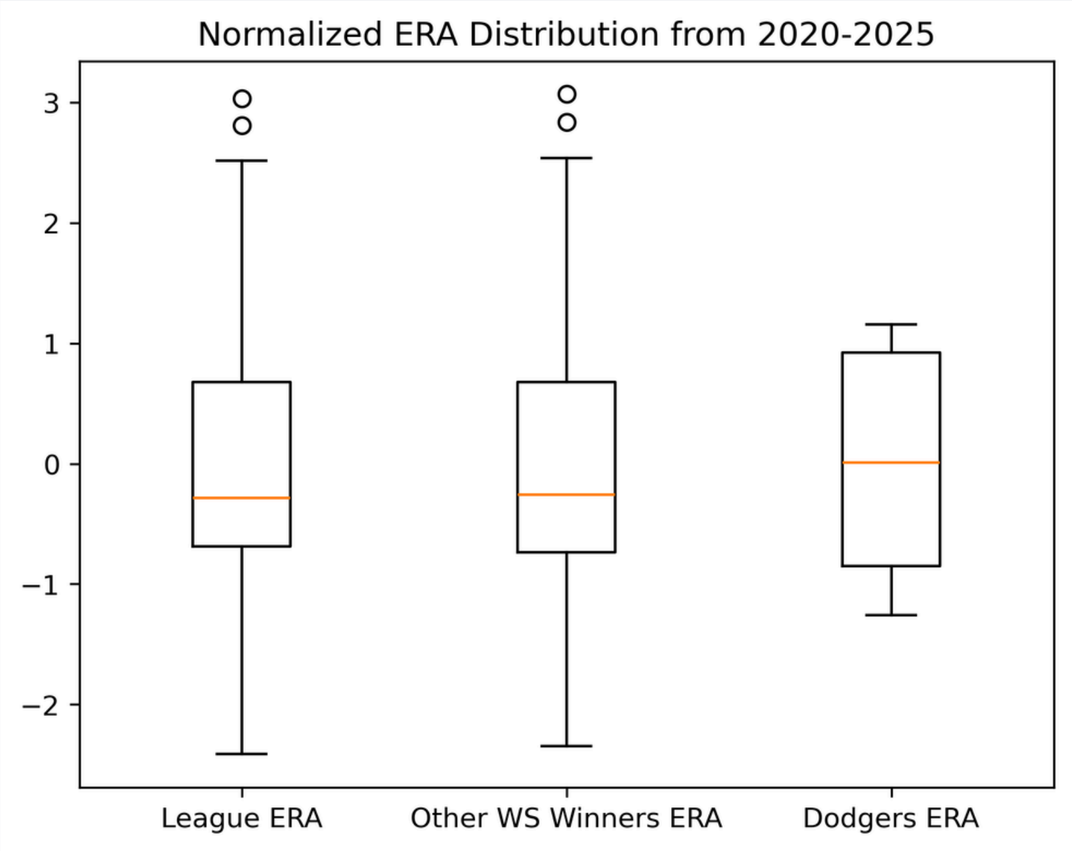
Dodgers vs League ERA

- Dodgers consistently have a lower ERA than the rest of the league
- 2024-2025 Dodgers have higher ERA compared to other WS winners however they won the WS both years
- Dodgers ERA spike is due to injuries



Normalized ERA Distribution

- League ERA: Made up of 29 teams, a wider spread because of different skill levels
- Other WS ERA: Very similar to the league, suggesting that ERA may not be the sole factor in going to the WS
- Dodgers ERA: Very little spread, showing consistency; however, their typical performance is worse (higher median)



Difference in Means Test: ERA and Winning

The null hypothesis states there is no difference in winning percentage between teams with high ERA and low ERA. The alternative hypothesis states there IS a difference

WE REJECT THE NULL HYPOTHESIS

- Compared winning percentage of low vs high ERA teams
- Used median ERA to split teams into two groups
- Conducted a t-test
- P-value essentially = 0
- Reject the null hypothesis
- Low ERA teams win significantly more

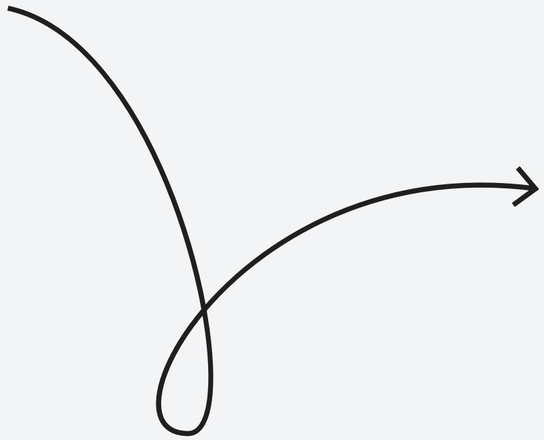
Regression Analysis: Stats and Winning Percentage

- Ran regressions with WinPct as dependent variable
- BA → weak positive relationship
- OPS → stronger positive relationship
- ERA → strong negative relationship
- ERA shows strongest overall relationship

Conclusion

Overall, BA shows the weakest relationship with winning, OPS shows a stronger offensive relationship, and ERA shows the strongest relationship overall. This suggests that pitching performance may be the most important factor in determining team success.

Thank You



Questions?

Sources

- 2020 Stats: https://www.baseball-reference.com/leagues/majors/2020.shtml
- 2021 Stats: https://www.baseball-reference.com/leagues/majors/2021.shtml
- 2022 Stats: https://www.baseball-reference.com/leagues/majors/2022.shtml
- 2023 Stats: https://www.baseball-reference.com/leagues/majors/2023.shtml
- 2024 Stats: https://www.baseball-reference.com/leagues/majors/2024.shtml
- 2025 Stats: https://www.baseball-reference.com/leagues/majors/2025.shtml